Module Review: Pyoverdine non-proteinogenic precursor supply in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Local bucket reviewed:** `kegg:ppu00975` — "Biosynthesis of various siderophores" (module area: other_kegg_pathway) **Module scope:** cytoplasmic supply of two non-proteinogenic amino-acid building blocks to the pyoverdine NRPS assembly line — N5-hydroxy-L-ornithine (via PvdA) and L-2,4-diaminobutyrate (via PvdH).

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1. Executive summary

The precursor-supply module has **exactly two enzymatic steps**, and **both are satisfiable in KT2440**:

Module step	Reaction	KT2440 gene	Status
1. N5-hydroxy-L-ornithine supply	L-ornithine + O_2 + NADPH → N5-hydroxy-L-ornithine	**pvdA** / **PP_3796** (Q88GC8)	**covered** — but **missing from the KEGG bucket & candidate list (gap in local metadata)**
2. L-2,4-diaminobutyrate supply	L-aspartate-4-semialdehyde + L-Glu ⇌ L-2,4-diaminobutyrate + 2-oxoglutarate	**pvdH** / **PP_4223** (Q88F75)	**covered** — high confidence

Two of the three candidate genes in the bucket are problematic for *this* module: - **PP_2800** (Q88J49) is a **likely over-propagated annotation** — a symbol-less class-III DABA-aminotransferase located in a polyamine/GABA gene neighborhood ~1.5 Mbp away from the pyoverdine cluster. It is **not** the pyoverdine PvdH and should not be counted toward module satisfiability. - **PvdY / PP_4245** (Q88F54) is an **acyl/acetyltransferase that is out of scope** for precursor supply; it belongs to pyoverdine side-chain tailoring, and its EC 2.3.1.102 mapping is a broad KO transfer of uncertain accuracy.

Bottom line: the module is **fully covered** in KT2440, but the *local bucket boundaries are wrong*: the true step-1 gene (pvdA/PP_3796) is absent, one bucket gene (PP_2800) is over-propagated noise, and one (PvdY) is out-of-scope. The bucket needs curation, not the biology.

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2. Target-organism pathway definition

Included biochemical process (this module only): cytoplasmic biosynthesis of the two unusual amino-acid precursors that the pyoverdine non-ribosomal peptide synthetases (NRPS) load onto the assembly line: 1. **N5-hydroxy-L-ornithine** — the hydroxamate iron-chelating precursor, made by the FAD/NADPH-dependent L-ornithine N5-monooxygenase **PvdA**. 2. **L-2,4-diaminobutyrate (Dab)** — a chromophore/backbone precursor (the fluorescent dihydroxyquinoline chromophore is a condensation product of D-tyrosine and L-2,4-diaminobutyrate;

P 15317763),

made by the PLP-dependent transaminase **PvdH** from L-aspartate-4-semialdehyde.

Explicitly out of scope (keep separate): NRPS peptide assembly (PP_4221/PP_4243 ferribactin synthase subunits, PvdL/PvdI/PvdJ/PvdD), N5-hydroxyornithine *tailoring* (formylation/acylation, e.g. PvdF), periplasmic maturation (PvdP/PvdM/PvdN/PvdO/PvdQ), ferribactin/pyoverdine export, ferripyoverdine uptake (FpvA), and iron release.

Neighboring maps to keep separate: - KEGG **ppu00260** "Glycine, serine and threonine metabolism" — PP_2800 is dual-mapped here; this is its more plausible home. - **Ectoine biosynthesis** and **1,3-diaminopropane / polyamine (putrescine/spermidine) metabolism** — these also use class-III DABA/2,4-diaminobutyrate aminotransferases (K00836, EC 2.6.1.76) and are the likely true context of PP_2800. - The broad KEGG overview map **ppu00975 "Biosynthesis of various siderophores"** is a *multi-siderophore* aggregate map (aerobactin, staphyloferrin, ferrioxamine, petrobactin, putrebactin, etc. appear in its compound list). It is **not** a pyoverdine-specific module; membership by KO/EC alone does not imply a pyoverdine role.

Alternate names / database definitions: pyoverdine = pyoverdin = PVD; PvdH = "diaminobutyrate-2-oxoglutarate transaminase" (EC 2.6.1.76, KO K00836); PvdA = "L-ornithine N5-monooxygenase / ornithine 5-monooxygenase / ornithine hydroxylase" (KO K10531, EC 1.14.13.195/1.14.13.196; UniProt lists EC 1.13.12.-).

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3. Expected step model

#	Step	Enzyme family	Expected reaction	Expected in KT2440?
1	Ornithine N5-hydroxylation	SidA/PvdA class-B flavin-dependent monooxygenase	L-Orn → N5-OH-L-Orn	Yes (pyoverdine producer)
2	Dab formation	Class-III PLP aminotransferase (PvdH type)	L-Asp-4-semialdehyde ⇌ L-2,4-Dab	Yes

Upstream context (not part of module but required for flux): **Asd** (aspartate-β-semialdehyde dehydrogenase) supplies L-aspartate-4-semialdehyde for step 2; *asd* knockouts also abolish pyoverdine (P 15317763). Asd is a housekeeping gene (aspartate-family amino-acid biosynthesis) and is present in KT2440, but is correctly outside this module.

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4. Candidate genes and evidence

pvdA / PP_3796 (Q88GC8) — NOT in bucket, but the true step-1 gene ★ promote

- **Role:** L-ornithine N5-monooxygenase → N5-hydroxy-L-ornithine (module step 1).
- **Evidence type:** KT2440 = homology/ortholog assignment (UniProt gene name *pvdA*; KEGG KO **K10531**, EC 1.14.13.195/196; SidA/PvdA family). Reaction chemistry is **directly established for the P. aeruginosa ortholog**: purified PvdA is a FAD/NADPH flavin-

monooxygenase catalyzing N5-hydroxylation of L-ornithine, pH optimum 8.0, Km(Orn) ≈ 0.58 mM, tightly coupled to hydroxylamine formation
(P 17015659 ; mechanism
P 17900176 ; family kinetics
P 20650894).

- **Species transfer:** strong (conserved gene name, KO, EC, family).
- **Caveat:** absent from the KEGG ppu00975 gene list and from local candidate metadata — a **bucket gap**, not a biological gap. Located at ~4.32 Mb, separate from the main NRPS cluster (a common arrangement for *pvdA*).

pvdH / PP_4223 (Q88F75) — high-confidence step-2 gene

- **Role:** diaminobutyrate-2-oxoglutarate transaminase → L-2,4-diaminobutyrate (module step 2).
- **Evidence type:** KT2440 = ortholog assignment (gene name *pvdH*, EC 2.6.1.76, KO K00836, class-III PLP aminotransferase) **plus genomic embedding in the pyoverdine cluster** (immediately flanked by PP_4221/PP_4222 ferribactin-synthase/SyrP-type genes). The P. aeruginosa PAO1 ortholog is **directly characterized**: interconverts aspartate-β-semialdehyde and L-2,4-diaminobutyrate, ping-pong mechanism, highest specificity for α-ketoglutarate (41× over pyruvate); *pvdH* knockouts cannot make pyoverdine without exogenous Dab, and PvdH homologues cluster within pyoverdine loci across *Pseudomonas* (P 15317763).
- **Species transfer:** strong.
- **Caveat:** none major; the EC/KO is shared with unrelated DABA-aminotransferases, so annotation should be anchored on the cluster location, not EC alone.

PP_2800 (Q88J49) — likely over-propagation, exclude from module

- **Role (as annotated):** "putative diaminobutyrate-2-oxoglutarate transaminase" (K00836, EC 2.6.1.76); **no gene symbol**; UniProt assigns no EC.
- **Why it is in the bucket:** shares KO/EC with PvdH, so KEGG lists it in the multi-siderophore map ppu00975 (and simultaneously in ppu00260 Gly/Ser/Thr metabolism).

- **Evidence against a pyoverdine role:** located at ~3.19 Mb, **far from the pyoverdine cluster**; its genomic neighborhood is polyamine/GABA-related — PP_2799 (class-III aminotransferase), PP_2801 (γ-aminobutyraldehyde dehydrogenase), PP_2802 (amino-acid permease). This is the signature of a general DABA/polyamine aminotransferase, not a pyoverdine enzyme. **KT2440 encodes only two K00836 DABA-aminotransferases (PP_2800 and pvdH/PP_4223) and lacks the ectoine pathway entirely (no ectA/K06718, no ectC/K06720), so PP_2800 is not an ectoine EctB** — its most plausible role is 1,3-diaminopropane/polyamine metabolism.
- **Species transfer:** N/A (this is KT2440-intrinsic genomic-context evidence).
- **Caveat:** do **not** count toward module satisfiability; mark `candidate_uncertain` and promote to full review to nail down its true polyamine/ectoine role.

pvdY / PP_4245 (Q88F54) — out of module scope, ambiguous annotation

- **Role (as annotated):** "hydroxyproline acetylase" EC 2.3.1.- (UniProt); KEGG maps to **K03896** "acetyl-CoA:N6-hydroxylysine acetyltransferase" EC 2.3.1.102 (an aerobactin/IucB-type activity).
- **Evidence type:** genomic embedding in the pyoverdine cluster (~4.83 Mb, near PP_4243 NRPS subunit and PP_4244 alternative sigma factor) supports a pyoverdine association, but **no direct functional characterization of PvdY was retrieved**, and the two annotations (hydroxyproline acetylase vs N6-hydroxylysine acetyltransferase) are inconsistent.
- **Relevance to this module:** PvdY is an **acyl/acetyltransferase tailoring enzyme**, not a supplier of N5-OH-ornithine or Dab. It does not satisfy either defined module step.
- **Caveat:** its EC 2.3.1.102/K03896 mapping is a broad cross-siderophore KO transfer of uncertain accuracy; promote to full review.

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5. Gaps, ambiguities, and likely over-annotations

- **Gap in local metadata (not in biology):** step-1 gene **pvdA/PP_3796 is absent** from the bucket and candidate list. This is the single most important curation fix — add PP_3796 as the step-1 gene.
- **Over-propagation: PP_2800** is a class-III aminotransferase co-classified into ppu00975 purely via EC 2.6.1.76/K00836. Genomic context (polyamine/GABA cluster, distal to pvd genes) argues it is not a pyoverdine enzyme. KT2440 has only two K00836 genes (PP_2800,

pvdH) and **no ectoine pathway** (ectA/ectC absent), ruling out an EctB role and pointing to 1,3-diaminopropane/polyamine metabolism. Classic KO/EC over-propagation into an aggregate map.

- **Out-of-scope bucket member: PvdY/PP_4245** is a pyoverdine-cluster acetyltransferase but not a precursor-supply enzyme; it inflates the apparent bucket while not covering any module step.
- **Ambiguous EC/KO mappings to flag:**
- EC 2.6.1.76 / K00836 is shared by pyoverdine PvdH, ectoine EctB, and polyamine DABA-aminotransferases — EC/KO alone cannot assign a pyoverdine role; use cluster location.
- PvdY: EC 2.3.1.- vs EC 2.3.1.102 (K03896) inconsistency; likely over-broad.
- PvdA: UniProt EC 1.13.12.- vs KEGG EC 1.14.13.195/196 — reconcile to the ornithine-N5-monooxygenase EC.
- **Not-expected steps:** none of the two module steps are missing biologically; there is no evidence that KT2440 replaces PvdA or PvdH with a lineage-specific alternative. (KT2440 is a well-established pyoverdine producer.)

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6. Module and GO-curation recommendations

Per-step module status: - **Step 1 (N5-hydroxy-L-ornithine) →** `covered`, by **pvdA/PP_3796** (add this gene to the bucket/module; it is currently a metadata gap). - **Step 2 (L-2,4-diaminobutyrate) →** `covered`, by **pvdH/PP_4223** (keep).

Bucket/candidate corrections: - **PP_2800 →** `candidate_uncertain` **and remove from pyoverdine module** (reassign to polyamine/ectoine amino-acid metabolism; retain in ppu00260 context). Likely over-annotation. - **PvdY/PP_4245 → out_of_scope for this module** (tailoring, not precursor supply). If module boundaries are later widened to include N5-hydroxyornithine acylation, revisit — but as defined, `module_needs_revision` only in the sense that PvdY should be moved out.

Module boundary verdict: the *generic* module boundaries (PvdA + PvdH) are **biologically correct for KT2440**; the *local KEGG bucket* boundaries are wrong (missing pvdA; includes over-propagated PP_2800 and out-of-scope PvdY). Recommend decoupling the pyoverdine precursor-supply module from raw `kegg:ppu00975` membership.

GO-curation notes: - pvdA/PP_3796: GO:0008442? no — appropriate terms: **GO:0030410 (nicotianamine? no)** → use **GO:0004497/flavin monooxygenase** parent with **"L-ornithine N5-monooxygenase activity"** (MF) and **GO:0019538/siderophore biosynthesis (BP; GO:0019290 siderophore biosynthetic process)**. Confirm an ornithine-N5-monooxygenase-specific MF term exists; request one if absent. - pvdH/PP_4223: MF **"diaminobutyrate-2-oxoglutarate transaminase activity"** (EC 2.6.1.76); BP **GO:0019290 siderophore biosynthetic process / pyoverdine biosynthesis**. - Flag EC/KO-driven annotations (PP_2800, PvdY) as electronic/over-propagated (IEA), needing experimental or context-based downgrade.

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7. Genes to promote to full `fetch-gene` review

1. **pvdA / PP_3796 (Q88GC8)** — highest priority; it is the missing step-1 gene and must be added to the module.
2. **PP_2800 (Q88J49)** — resolve its true role (polyamine/1,3-diaminopropane vs ectoine DABA-aminotransferase); confirm removal from pyoverdine module.
3. **pvdY / PP_4245 (Q88F54)** — reconcile conflicting acetyltransferase annotations and confirm out-of-scope status.

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8. Key references

- Vandenende, Vlasschaert, Seah (2004). *Functional characterization of an aminotransferase required for pyoverdine siderophore biosynthesis in Pseudomonas aeruginosa PAO1.* J Bacteriol.

  P 15317763

  — Direct: PvdH makes L-2,4-diaminobutyrate; chromophore = D-Tyr + Dab; PvdH homologues sit in pyoverdine loci across *Pseudomonas*.
- Ge & Seah (2006). *Heterologous expression, purification, and characterization of an L-ornithine N5-hydroxylase involved in pyoverdine siderophore biosynthesis in P. aeruginosa.* J Bacteriol.

  P 17015659

  — Direct: PvdA N5-hydroxylates L-ornithine (FAD/NADPH).

- Meneely & Lamb (2007). *Biochemical characterization of a FAD-dependent monooxygenase, ornithine hydroxylase from P. aeruginosa.* Biochemistry.

  P 17900176

  — PvdA mechanism.
- Mayfield et al. (2010). *Comprehensive spectroscopic, steady state, and transient kinetic studies of a representative siderophore-associated flavin monooxygenase.* Biochemistry.

  P 20650894

  — SidA/PvdA family kinetics.
- Koch et al. (2010). *The acylase PvdQ has a conserved function among fluorescent Pseudomonas spp.* — confirms pyoverdine biosynthesis machinery is conserved and functional in P. putida KT2440.

  P 23766117
- Databases: KEGG ppu00975 / ppu:PP_2800 / ppu:PP_3796 / ppu:PP_4223 / ppu:PP_4245; UniProt Q88J49, Q88GC8, Q88F75, Q88F54 (accessed 2026-08-31).

Evidence provenance: Reaction chemistry for both module steps is **direct experimental** evidence from *P. aeruginosa* orthologs (strong transfer to KT2440). KT2440-specific claims (gene identities, KO/EC, chromosomal positions, gene neighborhoods) are from KEGG/UniProt for strain KT2440 and are **direct genomic** evidence. The over-propagation and out-of-scope calls rest on KT2440 genomic-context reasoning, not on direct functional assays of PP_2800/PvdY — these remain the principal open questions.
